



GalaxEye Space Technical Assignment Report

Project Title :

EO-SAR Binary Change Detection using U-Net

Name : Penukonda Vari Deepak

Qualification : Bachelor of Engineering (B.E.)

In

**Computer Science and Engineering (Artificial Intelligence and
Data Science)**

1. Abstract

This project focuses on binary change detection using multimodal Earth Observation (EO) and Synthetic Aperture Radar (SAR) imagery for post-disaster building damage assessment. The objective was to develop an end-to-end deep learning pipeline capable of identifying damaged regions from paired EO-SAR image data through semantic segmentation. The original multi-class damage labels were remapped into binary classes to simplify the problem into damaged versus non-damaged region detection. A U-Net architecture with a ResNet34 encoder backbone was implemented using fused EO RGB and SAR inputs as a four-channel representation. The pipeline included TIFF image preprocessing, binary mask generation, dataset loading, model training, evaluation, and qualitative visualization. During experimentation, severe class imbalance emerged as the primary challenge, causing the model to initially collapse toward predicting the dominant background class. To address this issue, Weighted Binary Cross Entropy combined with Dice Loss was implemented to improve sensitivity toward sparse damaged regions. Threshold tuning was additionally performed to improve recall performance. The final model was evaluated using IoU, Precision, Recall, F1 Score, and confusion matrix analysis on validation and test datasets. Qualitative prediction analysis demonstrated successful detection of several damaged structures while also revealing limitations caused by SAR noise, low-contrast regions, and sparse damage patterns. Overall, the project demonstrates a complete multimodal EO-SAR segmentation workflow and establishes a strong baseline for future transformer-based and attention-driven remote sensing change detection research.

2. Literature Survey

Remote sensing change detection is widely used for disaster monitoring, urban analysis, and infrastructure assessment. Traditional methods such as image differencing and thresholding relied on handcrafted techniques but struggled with complex multimodal data. Deep learning models, especially convolutional neural networks, significantly improved segmentation accuracy in remote sensing tasks. U-Net became one of the most widely used architectures due to its encoder-decoder structure and skip connections that preserve spatial information. Recent research has focused on multimodal fusion using Earth Observation (EO) optical imagery and Synthetic Aperture Radar (SAR) imagery. EO imagery provides rich spectral features, while SAR imagery offers structural information and robustness under poor weather conditions. However, SAR data introduces speckle noise and modality differences that complicate feature learning. Transformer-based architectures such as ChangeFormer and attention-based segmentation networks have recently shown strong

performance in EO-SAR change detection tasks. Another major challenge is severe class imbalance, where damaged regions are significantly fewer than background pixels. To address this issue, this project implemented a U-Net architecture with Weighted BCE and Dice Loss for improved damaged region detection and segmentation performance.

3. Methodology.

The project was developed as an end-to-end binary change detection pipeline using multimodal EO and SAR imagery. The original four-class damage labels were remapped into binary classes where labels 0 and 1 represented background, while labels 2 and 3 represented damaged regions. The preprocessing stage included TIFF image loading, EO RGB extraction, SAR integration, resizing, normalization, and binary mask generation. EO RGB channels and SAR imagery were fused into a four-channel input representation for multimodal learning. A custom PyTorch Dataset and DataLoader pipeline was implemented for efficient batch processing. A U-Net architecture with a ResNet34 encoder backbone was selected because of its strong segmentation performance and spatial feature preservation. The model was trained using the AdamW optimizer with a learning rate of $1e-4$ and batch size of 4. Severe class imbalance was identified as the primary challenge during experimentation. To address this issue, Weighted Binary Cross Entropy combined with Dice Loss was implemented to improve sensitivity toward sparse damaged regions. The final model was evaluated using IoU, Precision, Recall, F1 Score, confusion matrix analysis, and qualitative prediction visualization.

4. Result

The final model was evaluated on both validation and test datasets using standard semantic segmentation metrics including Intersection over Union (IoU), Precision, Recall, F1 Score, and confusion matrix analysis. Initial experiments using standard Binary Cross Entropy loss resulted in prediction collapse toward the majority background class because of severe class imbalance within the dataset. After implementing Weighted BCE combined with Dice Loss and threshold tuning, the model demonstrated improved sensitivity toward damaged regions, although overall segmentation performance remained limited due to sparse positive samples and SAR noise complexity.

Validation Results

Metric	Value
IoU	0.209

Precision	0.3009
Recall	0.2914
F1 Score	0.1988

Test Results

Metric	Value
IoU	0.2339
Precision	0.3009
Recall	0.3550
F1 Score	0.2220

Confusion Matrix

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[[5006216  1935]
 [ 38108    13]]
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The confusion matrix indicates that the model strongly favored background predictions, resulting in very low true positive detections for damaged regions. This behavior was primarily caused by severe class imbalance, where background pixels significantly outnumbered damaged pixels. Qualitative visualization analysis was performed using multiple prediction examples covering both successful detections and failure cases. The model showed better performance on large damaged structures with clear boundaries, while struggling with sparse damage regions, low-contrast structures, and SAR-specific speckle noise. Failure cases revealed fragmented predictions and missed detections, highlighting the limitations of the current baseline architecture under highly imbalanced multimodal remote sensing conditions.

5. Future Work

The current U-Net baseline establishes a functional end-to-end EO-SAR change detection pipeline, but several improvements can be explored to improve segmentation performance and multimodal feature learning. Future work would focus on transformer-based architectures such as ChangeFormer and Attention U-Net to better capture long-range spatial dependencies and improve

EO-SAR feature fusion. Since severe class imbalance remained the primary challenge, advanced loss functions including Focal Loss and Tversky Loss would be investigated to improve sensitivity toward sparse damaged regions. Stronger augmentation strategies, multi-scale training, and patch-based learning could further improve generalization and small-object detection performance. Dedicated SAR denoising and preprocessing techniques would also be explored to reduce speckle noise and improve structural feature extraction. Self-supervised pretraining on large remote sensing datasets may additionally improve feature representation quality under limited labeled data conditions. Future experimentation would also include systematic hyperparameter tuning, threshold optimization, and ablation studies to better understand failure cases and optimize segmentation robustness in multimodal EO-SAR environments.

6. Conclusion

This project successfully implemented an end-to-end binary change detection pipeline using multimodal EO and SAR imagery for post-disaster building damage assessment. The workflow included data preprocessing, binary label remapping, multimodal feature fusion, semantic segmentation, evaluation, and qualitative visualization. A U-Net architecture with a ResNet34 encoder was used as the baseline segmentation model, while Weighted BCE combined with Dice Loss was implemented to address severe class imbalance within the dataset. The experiments demonstrated that class imbalance and SAR noise were the dominant challenges affecting segmentation performance. Although the final model was capable of detecting several damaged regions, overall IoU and F1 performance remained limited because the model still favored majority background predictions. The results highlight the complexity of EO-SAR multimodal fusion and the difficulty of detecting sparse damaged structures under noisy remote sensing conditions. Despite these limitations, the project establishes a functional research baseline and demonstrates strong engineering understanding of preprocessing, imbalance handling, multimodal segmentation, and evaluation workflows. Future improvements involving transformer-based fusion architectures, advanced loss functions, stronger augmentation strategies, and SAR-specific preprocessing are expected to significantly improve segmentation robustness and damaged region detection accuracy.